Equipment

ONR FY2027 YIP, NOFO N0001426SF004. PI: Sterling G. Baird, Brigham Young University (BYU)

Equipment available. The equipment below is in place and available to the project at no cost, except where a facility charges hourly rates, which are in the budget.

- AMAZEMET rePOWDER ultrasonic atomizer with an induction crucible module (to 1,300 °C), with the coated plates and boron nitride nozzles that the manufacturer supplies for copper alloys (PI laboratory, BYU; purchased with start-up funds in 2026).
- Liquid-handling gantry (CubXL, Ursa Laboratories) with a pipette, a vial capper and two cameras, run by our own scripts (PI laboratory). It runs the electrochemical corrosion screen and samples the immersion solutions.
- Powder handling, sieving and specimen preparation equipment in the PI laboratory: an open-source auger powder doser built in the laboratory, a sieve stack from 45 to 250 μm with a sieve shaker, a Hall flowmeter funnel, an analytical balance, glove bags, dry powder storage and sectioning and mounting equipment.
- Scanning electron microscopes with EDS and EBSD (Apreo C, Helios 600 with a focused ion beam, Verios) and a Tecnai TF-20 transmission electron microscope at the BYU Microscopy Center.
- Agilent 8900 ICP-MS/MS and XPS at the BYU Spectroscopic Facility, and X-ray diffraction at the BYU X-ray Diffraction Facility.
- Metallographic polishers (PX500 automatic, PX300 manual and vibratory), a Keyence VHX-7000 digital optical microscope, and CNC machining at BYU.
- Instron load frames in the materials testing laboratory of the Department of Mechanical Engineering, for sub-size tensile tests [TODO Sterling: confirm].
- The BYU Office of Research Computing cluster (about 28,000 CPU cores and 367 GPUs).
- Blown-powder DED through purchased sessions: the InssTek MX-Lab (500 W, six powder feeders) at the University of Utah for tracks and small deposits from up to 0.5 kg of powder, and the FormAlloy X2 at the Colorado School of Mines for 2 kg builds and repair coupons.

Equipment to be purchased with project funds. One potentiostat with corrosion software and a flat cell, Year 1, $18,000. This is a budgetary estimate for a research-grade single-channel instrument with electrochemical impedance capability and corrosion software (for example, a Gamry Interface 1010E or a BioLogic SP-150e, each with a flat cell and reference electrodes). A vendor quote accompanies the budget. The electrochemical corrosion screen (open-circuit potential, linear polarization resistance, potentiodynamic polarization and impedance in substitute ocean water) is one step of the workflow that the cost-aware search schedules. The instrument therefore has to be available every day and run by our own scripts on the gantry described above. No potentiostat at BYU can be dedicated to the project in this way. The shared instruments on campus are booked by the hour, and the search cannot schedule them. Equipment is excluded from the modified total direct cost base.

Government-owned equipment. None.